Type MS · Explain how vaccination or infection gives active immunity

Answer 1 of 7 · Question 4 part c · 5 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

number				
4	(a) (i)	A (bacterium); B doesn't cause it C doesn't cause it D doesn't cause it		1
	(ii)	• no of cases decreases as vaccination rate increases; • lag in effect; • fluctuations in number of cases; • despite percentage of vaccinations remaining constant;		4
	(b)	2.75 per 100 000; $\frac{2.75 \times 3\,450\,000}{100\,000}$; = 95 people;	correct answer without working = 3 ECF = 2	3
	(c)	• inject weakened/attenuated organism; • causes antibody production; • antibodies remain in blood; • memory cells formed; • on infection antibodies can respond quickly;		5

Total 13 marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 1 of 7 · Question 4 part c · 5 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 2 of 7 · Question 8 part c · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

8 (a)	Three from: • (parasite) carried by mosquito/mosquito is a vector; • mosquito bites human; • parasite/infected blood drawn/sucked up into mosquito; • infected blood transmitted to other people;		Max 3
(b)(i)	• fewer deaths caused by *P.vivax* in R than Q • (but) more deaths caused by *P.vivax* in R than in Q as a proportion of the total; • 190 000 fewer deaths caused by *P.vivax* in R/300 less deaths in Q caused by *P.vivax*; • 0.5% deaths caused by *P.vivax* in Q/62.5% of deaths caused by *P.vivax* in R;		1 1 1
(ii)	more mosquitoes in one region than another/climate favours breeding of mosquitoes/more dense population of people/better health care/use of (named) preventative measures;		1
(c)	Three from: Fewer people with malaria/reduced incidence of malaria; Immunity/antibodies against parasite/herd immunity; Reduction in population of/death of parasite; Less transmission (from one person to another);		Max 3
(d)	Three from: (sexual reproduction) • gives rise to variation in offspring; • variation provides a survival advantage; • parasite more likely to survive in a changing environment; (asexual reproduction) • parasite can reproduce faster; • no need to find a mate/only one parent needed; • larger number of offspring produced;	Answer must contain at least one advantage of each method	Max 3

Type MS · Explain how vaccination or infection gives active immunity

Answer 2 of 7 · Question 8 part c · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 3 of 7 · Question 9 part c · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

9 (a)	Any three of the following • causes scarring/cirrhosis; • unable to detoxify substances / produce bile; • gene mutation; • leading to cancer;		3
(b)	Any six from the following • virus attaches to (host) cell; • DNA injected into cell; • DNA becomes incorporated into host cell DNA; • viral DNA replicated along with host cell DNA; • protein synthesis; • new virus produced; • lysis of host cell; • new viruses released (to infect new cells);		6
(c)	• antigens injected; • white cells/lymphocytes produce antibodies; • memory cells produce antibodies faster;		3
			Total 12

Type MS · Explain how vaccination or infection gives active immunity

Answer 3 of 7 · Question 9 part c · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain how vaccination or infection gives active immunity

Answer 4 of 7 · Question 3 part b.iii · 2 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
3	(a)	pathogens;		1
	(b) (i)	1.2 years;		1
	(ii)	• active; • artificial;		2
	(iii)	Any two from • weakened/dead/attenuated organisms; • toxins; • antigens; • RNA;	Allow named bacterium/virus Ignore genetic material	2
	(c)	Any two from • passed from mother; • during breastfeeding/from breast milk; • across placenta;	Ignore umbilical cord	2
	(d)	Any five from • antibody level falls; • below safe level/to unsafe level; • greater risk of disease; • booster dose increases antibody levels/remains above unsafe level; • person immune/can't catch disease; • increased number of memory cells;	Allow faster response	5
	(e)	Any two from • antibody level higher; • for longer period/decreases more slowly/remain for many years; • levels of antibody increase faster after booster;		2

Total 15 marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 4 of 7 · Question 3 part b.iii · 2 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 5 of 7 · Question 3 part d · 5 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number				
3	(a)	pathogens;		1
	(b) (i)	1.2 years;		1
	(ii)	• active; • artificial;		2
	(iii)	Any two from • weakened/dead/attenuated organisms; • toxins; • antigens; • RNA;	Allow named bacterium/virus Ignore genetic material	2
	(c)	Any two from • passed from mother; • during breastfeeding/from breast milk; • across placenta;	Ignore umbilical cord	2
	(d)	Any five from • antibody level falls; • below safe level/to unsafe level; • greater risk of disease; • booster dose increases antibody levels/remains above unsafe level; • person immune/can't catch disease; • increased number of memory cells;	Allow faster response	5
	(e)	Any two from • antibody level higher; • for longer period/decreases more slowly/remain for many years; • levels of antibody increase faster after booster;		2

Total 15 marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 5 of 7 · Question 3 part d · 5 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 6 of 7 · Question 1 part d · 1 mark

2024 JUN · Paper 02 · find part d in this mark scheme extract

Question number				
1	(a)	<table><tr><th>Disease</th><th>Type of pathogen</th></tr><tr><td>cholera</td><td>Bacterium/bacteria;</td></tr><tr><td>AIDS</td><td>virus;</td></tr><tr><td>athlete's foot</td><td>Fungus/fungi;</td></tr></table>	Reject more than one line from any disease	(3)
	(b)	In the following order: bacterium; sexual intercourse; barrier; antibiotics;		(4)
	(c) (i)	• overall decrease; • fluctuates • steep decreases from 1990;		(2)
	(ii)	17 - 9; 8;	Allow tolerance of +-0.5 ecf for mp2	(2)
	(d)	C antigens,white blood cells produced,antibodies destroy pathogens; The following are incorrect because - A vaccinations do not contain antibodies B platelets are not produced following a vaccination D vaccinations do not produce platelets		(1)

Total for Question 1 = 12 marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 7 of 7 · Question 4 part b · 5 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number				
4	(a) (i)	method 1 = passive; method 2 = active;		2
	(ii)	• method 1 because antibodies injected; • immediate high level in blood/faster response; • to destroy, pathogen/disease organism;	Do not allow destroy / kill/attack disease	3
	(b)	• catch disease/pathogens enter body; • antibodies produced; • by white blood cells/lymphocytes; • memory cells produced; • rapid production of antibodies on re-infection/faster secondary immune response;		5

Total 10 marks

Type MS · Explain how vaccination or infection gives active immunity

Answer 7 of 7 · Question 4 part b · 5 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks